

Context Window

Earlier in the course, you met the calculator on your phone: type $2 + 2$ and you always get 4, because a programmer wrote every rule it follows. Same input, same answer, for everyone, every time. AI doesn't work that way. Consider the following.



Luke

Asks

"What car should I buy after I graduate from college?"

ChatGPT answers

"Great question, Luke. I definitely recommend a Jeep Cherokee."



Nate

Asks

"What car should I buy after I graduate from college?"

ChatGPT answers

"I'm happy you asked this question, Nate. At this point in your life, I recommend the Ford Raptor."

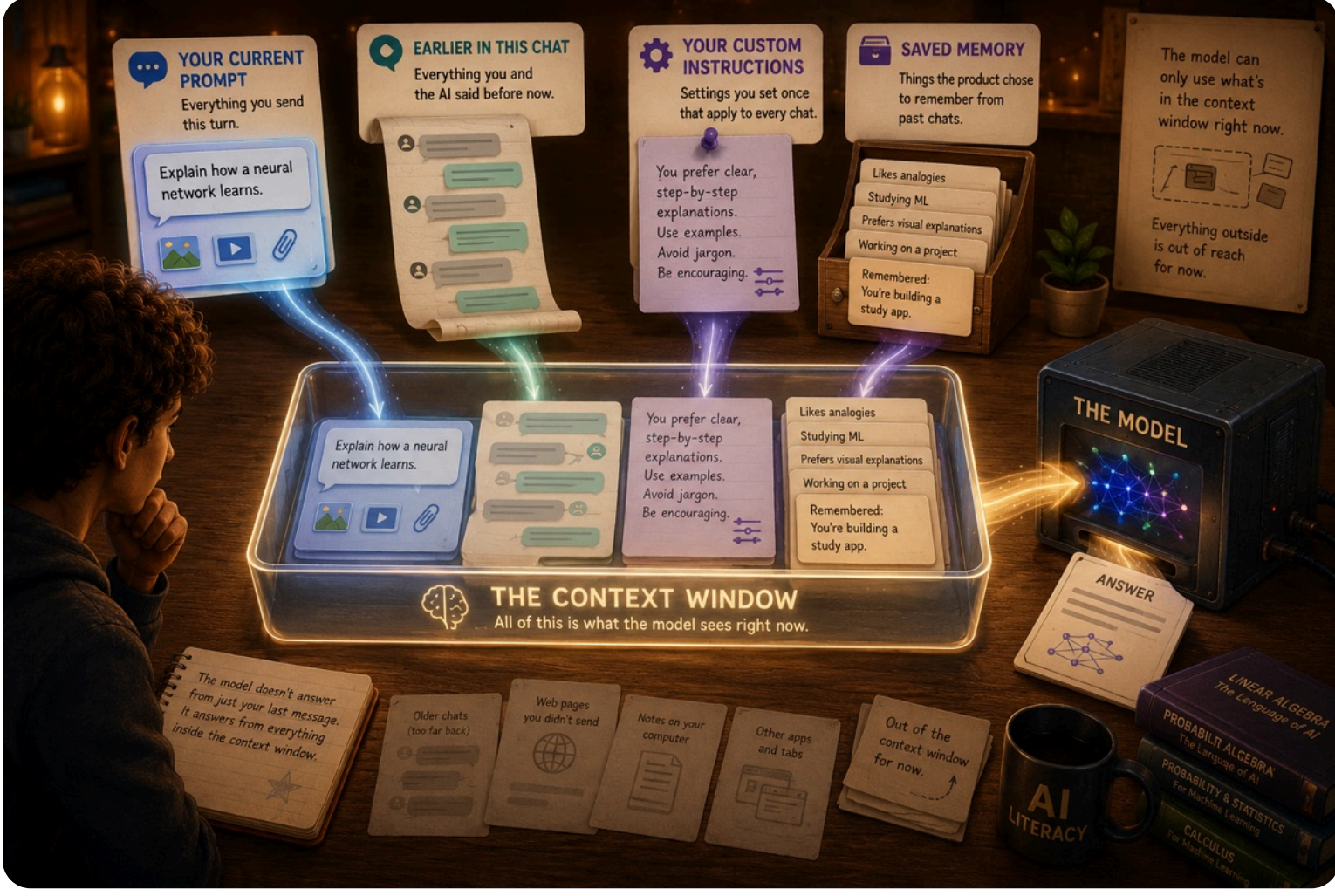
Wait a second. They're both using ChatGPT, but they got a different answer to the same question!

Some people think this is a reason not to trust AI. But the opposite is true. It's actually demonstrating the power of AI. It gave a different, more tailored answer to each.

Understanding why this happened is one of the keys to being smarter than the tool.

When you use a specific model of ChatGPT, it's the same tool everyone else starts with. But here's where it splits from the calculator: that calculator forgets you the instant it shows the answer, while ChatGPT knows more than the last question you typed. For example, earlier in his chat, Nate said he loves pickup trucks. When ChatGPT answers his question, it recognizes this and suggests the Ford Raptor.

What do you call everything the model can see when it answers your question? The **context window**. Think of it almost like the working memory the model uses to answer. And it builds that working memory from several places.



WHAT ACTUALLY GETS SENT

Here's the part that surprises people. When you hit send, AI doesn't just receive your last message. It receives the entire context window: your prompt, everything earlier in the chat, your custom instructions, and any saved memory. All of it, every single turn.

It feels like a conversation that remembers you. But the model treats every turn as a one-off: it wakes up, reads the entire window as if for the first time, answers, then starts fresh the next turn. It's the Chinese Room from Does AI Think? again: a fresh note slides under the door, the model answers from only what's written on it, and keeps nothing. The window is that note.

And everything you learned earlier applies to all of it. The whole window is broken into tokens, each token becomes a vector, and the entire thing runs through the layers together, every single time. That window is the only thing the model can see.

It's also why ChatGPT and Claude can feel slower as a chat gets long: there's simply more in the window to work through before the model can start answering.

CONTEXT WINDOW SIZE

The whole window is measured in tokens, the same chunks you met in the Tokens lesson. Pile up enough of them and you hit the context window's limit.

Before that limit, the model sees everything in the chat. After it, it doesn't. Some apps summarize the older parts. Others just forget.

And that limit is growing fast.

4,096 TOKENS

1st version of ChatGPT (November, 2022)

~3,000 words — About 6 pages

1 MILLION TOKENS

Latest Claude models

~1,000,000 words — About the size of the first 5 Harry Potter novels

Big or small, what fills the window is what shapes the answer. Same question, different windows. Your call:



You've seen what the model can see. Next: how it turns all of it into an answer, one word at a time.